Stage 4: Turning Points, Holiday Transmission, and Model Validation

Question 1. Why did “growth vs cumulative admissions” fail, and what did it show?

Step 1. Derive the line

$$\text{growth} = \gamma(R_0 s - 1),\ s = 1 - \kappa H/\bar{H}$$

$$\text{growth} = \gamma[R_0(1 - \kappa H/\bar{H}) - 1]$$

$$\text{growth} = \gamma(R_0 - 1) - \gamma R_0 \kappa (H/\bar{H}).$$

In $y = c + mx$, $x = H/\bar{H}$, the intercept is $c = \gamma(R_0 - 1)$ and the slope is $m = -\gamma R_0 \kappa$. With positive γ, R_0 and κ, the slope is negative: as cumulative admissions H increase, growth should fall if transmission R_0 remains fixed.

Step 2. Read the observations

Season	3 weeks before	2 weeks before	Last rising	First after
2023–24	0.32	0.43	0.32	−0.11
2024–25	0.52	0.58	0.33	−0.22
2025–26	0.64	0.58	0.13	−0.35

In 2023–24, growth rose by $0.43 - 0.32 = 0.11$. Counting only values strictly above 0.3, the pre-turn weeks were 3, 3 and 2 for the three seasons respectively. In 2025–26, the turn-week drop was $0.13 - (-0.35) = 0.48$ log-growth units.

Step 3. Compare theory and data

No consistent slow, steady decline appeared. Growth remained high or rose, then reversed abruptly. The observed points did not provide the reliable downward straight line needed to forecast a zero crossing.

Step 4. What else changes?

With $\gamma = 1/3$ fixed and s decreasing, the transmission quantity R_0 must rise sufficiently to sustain or increase growth. Possible reasons include more indoor mixing in colder months and school/seasonal social contacts. School closure over the winter break could abruptly reduce transmission. Here R_0 represents a time-varying transmission potential rather than a literally fixed basic reproduction number.

Step 5. Why no advance warning?

There were zero consistently declining multiweek sequences across all three seasons: 2023–24 rose before falling; 2024–25 and 2025–26 offered only two adjacent declining steps at most. A slope-based warning requires several stable pre-turn points; a reversal within a week gives little lead time.

Question 2. How does the holiday factor turn the epidemic?

Step 1. Reproduction during the break

$$R_holiday \approx R_t \times \text{factor} = 1.260 \times 0.8 = 1.008.$$

Since $1.008 > 1$, infections still grow very slowly. This is near-stalling, not an immediate decline.

Step 2. Exact threshold

1.260 f = 1 ⇒ f = 1/1.260 = 0.79365079… ≈ 0.794.

The selected factor 0.8 is bigger than the threshold, so it is not, by itself, sufficient to force R below 1 at these values. Additional susceptible depletion or a further transmission reduction could produce a turn.

Step 3. Seven-day lag and calendar

The break begins December 21, 2025. If infections begin declining that day, adding seven days gives December 28, 2025. The first full Sunday–Saturday reporting week starting on or after that date ends January 3, 2026. The actual national peak also ended January 3: it aligns with the hypothesized lagged turn, although R_holiday = 1.008 does not mathematically guarantee that turn.

Step 4. Return after the break

R_after ≈ 1.260 × 1 = 1.260 > 1.

If susceptibility remains close to s_now = 0.910, infections should rise again after January 4, followed by a mid-January admission rebound. The draft's Figure 7 and Section 6.4 show the blue forecast rebounding (about 46,000 weekly admissions), but the black observed series did not: admissions continued falling.

Step 5. Which seasons rebounded?

2024–25 rebounded after the January dip and peaked February 8. The seasons 2022–23 (December 3), 2023–24 (December 30), and 2025–26 (January 3) peaked near a holiday break without a later larger national peak. The model rebound is sometimes, not always, correct.

Question 3. Is the holiday effect real, or did we overfit?

Step 1. Count tuning and test seasons

Two settings were tuned: susceptible-depletion κ and holiday factor φ_H. Tuning used two seasons, 2023–24 and 2024–25; one season, 2025–26, was held out for testing.

Step 2. Convert WIS scores to percentage improvements

Improvement (%) = (1 − relative WIS) × 100.

Stage 4: (1 − 0.812) × 100 = 18.8%.

Stage 3: (1 − 0.997) × 100 = 0.3%.

Stage 4 near peak: (1 − 0.672) × 100 = 32.8%.

Adding holidays increased the overall improvement against baseline by 18.5 percentage points (18.8% versus 0.3%) and improved the near-peak score by 32.8% against baseline.

Step 3. Grid robustness

κ	2023–24 best	2024–25 best	2025–26 best
0.3	0.8 (0.783)	0.8 (0.747)	0.8 (0.812)
0.4	0.8 (0.728)	0.8 (0.963)	0.8 (0.703)
0.5	0.8 (0.709)	0.8 (1.296)	0.8 (0.635)

Factor 0.8 is best in 9/9 comparisons (100%). Consistency across seasons and κ choices weakens the case for a chance optimum, but cannot eliminate overfitting: only three historical seasons and factors {0.6, 0.7, 0.8, 1.0} were assessed. Values between 0.8 and 1.0 were not tested.

Step 4. Prospective 2026–27 predictions

Thanksgiving: Nov 26 + 7 days = Dec 3, 2026 → week ending Dec 5.

Winter break: Dec 21 + 7 days = Dec 28, 2026 → week ending Jan 2, 2027.

If the holiday effect is real, US admissions in the week ending December 5 should slow or decline compared with the preceding week, and growth should decrease in the week ending January 2. These are forecasts, not observations; a reduction in growth need not imply a decline in admissions.

Step 5. Limitations and checks

All four proposed factors merit consideration: travel can change mixing and admission location; fewer care visits can reduce observed admissions; late hospital reports can depress initial counts; cold weather can change transmission, though it does not inherently cause a drop. Compare districts or states with different school-break calendars, or examine whether school-aged children experience an earlier decline than adults. The fitted holiday factor captures a combined holiday-associated effect, not an isolated causal effect of school closures.